Part 5

Inferences about Population Variances
Confidence Interval for Population Variances
Estimation and Tests for Comparing Two Population Variances

INFERENCES ABOUT POPULATION VARIANCES

Estimation and Tests for a Single Population Variance

- Recall that the sample variance $s^2 = \sum(y - \bar{y})^2/(n-1)$ is the *point estimate* of σ^2.
- For tests and confidence intervals about σ^2 we use the fact that the sampling random variable

$$(n-1)\,S^2/\sigma^2 = \chi^2$$

has the *Chi-square Distribution* with $n-1$ degrees of freedom or *df*, when the sample is from a normal population.

- The chi-square distribution is nonsymmetric.
- Like the Student's t distribution, there is a different curve for each sample size n i.e., for each value of df.
- Percentiles of the chi-square distribution are given in Table 7.
- Plots of the chi-square distribution for $df = 5, 15$, and 30 are shown in Fig. 7.3
- The distribution appear to be more skewed for smaller values of df and become more symmetric as df increases.
- Because the chi-square distribution is nonsymmetric, the percentiles for probabilities at both ends needs to be tabulated.

Chi-square Distribution

FIGURE 7.3
Densities of the chi-square ($df = 5, 15, 30$) distribution

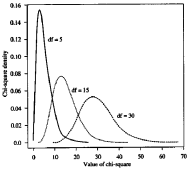


FIGURE 7.4
Critical values of the chi-square distribution with $df = 14$

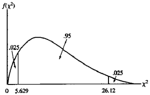
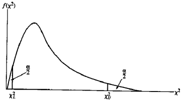


FIGURE 7.5
Upper-tail and lower-tail values of chi-square



A 100(1-α)% Confidence Interval for $\boldsymbol{\sigma^2}$

This has the form:

$$\frac{(n-1)s^2}{\chi^2_U} < \sigma^2 < \frac{(n-1)s^2}{\chi^2_L}$$

where

- Since $df = n - 1$ look up χ^2_{n-1} percentiles.
- χ^2_L is the lower-tail value with area $\alpha/2$ to the left.
- χ^2_U is the upper-tail value with area $\alpha/2$ to the right.
- The confidence interval for the standard deviation σ is found by taking square roots of both end points of above.

Example

Let's assume that the normal probability plot of the data appear to show that the sample is from a normal distribution.

From the data $n = 30$, $\bar{y} = 500.453$, $s = 3.433$ were calculated.

A 99% confidence interval for σ^2 is computed as follows:

Since $\alpha = .1, \quad \alpha/2 = .005$ and $1 - \alpha/2 = .995$, we compute

$\chi^2_L = \chi^2_{0.995,29} = 13.12 \quad \chi^2_U = \chi^2_{0.005,29} = 52.34$

Thus the endpoints for the required C.I. are, respectively,:

$$\frac{(n-1)s^2}{\chi^2_U} = \frac{29(3.433^2)}{52.34} = 6.53 \qquad \frac{(n-1)s^2}{\chi^2_L} = \frac{29(3.433^2)}{13.12} = 26.05$$

Thus $(6.53,\ 26.05)$ is a 99% C.I. for σ^2

Taking square roots of both endpoints of the interval for σ^2:

we get $(2.56,\ 5.10)$ as a 99% C.I. for σ.

Thus we are 99% confident that the standard deviation of the weights of coffee containers lies in the interval $(2.56,\ 5.10)$.

The filling machine was designed to be so that $\sigma = 4$ grams, and since this value falls in the above interval, the machine satisfies the design specification for standard deviation.

A separate confidence interval for μ is computed for checking whether the design specification for the mean is satisfied.

Tests of Hypotheses about σ^2

Test:

1. $H_0 : \sigma^2 \leq \sigma_0^2$ vs. $H_a : \sigma^2 > \sigma_0^2$
2. $H_0 : \sigma^2 \geq \sigma_0^2$ vs. $\sigma^2 < \sigma_0^2$
3. $H_0 : \sigma^2 = \sigma_0^2$ vs. $\sigma^2 \neq \sigma_0^2$

Test Statistic: $\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$

Rejection Region: for specified α and $df = n - 1$,

1. Reject H_0 if $\chi^2 > \chi_U^2$, where $\chi_U^2 \equiv \chi^2_{\alpha,\, n-1}$
2. Reject H_0 if $\chi^2 < \chi_L^2$, where $\chi_L^2 \equiv \chi^2_{1-\alpha,\, n-1}$
3. Reject H_0 if $\chi^2 > \chi_U^2$, where $\chi_U^2 \equiv \chi^2_{\alpha/2,\, n-1}$, or if $\chi^2 < \chi_L^2$, where $\chi_L^2 \equiv \chi^2_{1-\alpha/2,\, n-1}$

Handout - Example 7.2

Notes about Example

- The above inferences are based on the assumption of sampling from a normal population.
- They are more sensitive to departures from normality than the inferences about population mean.
- We don't have a CLT-type theorem which applies to S^2 like we have in case of $\bar{Y}$.
- Always plotting sample data as a preliminary procedure - using boxplot or normal probability plot to look for skewness, or outliers, is recommended.

Estimation and Tests for Comparing Two Population Variances

- We will consider the case of independent samples from two populations having variances σ_1^2 and σ_2^2.
- The main application is testing $\sigma_1^2 = \sigma_2^2$.
- Theory says that when the two populations are Normal, and sample sizes are n_1 and n_2 then the random variable

$$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}$$

has the F distribution with degrees of freedom $df_1 = n_1 - 1$ and $df_2 = n_2 - 1$.

Estimation and Tests for Comparing Two Population Variances - cont.

- df_1 is the degrees of freedom associated with S_1^2.
- df_2 is the degrees of freedom associated with S_2^2.
- The F distribution is a two parameter family indexed by df_1 and df_2.
- Some use the terms Numerator and Denominator degrees of freedom for df_1 and df_2.
- The F-statistic is calculated as

$$F_c = \frac{s_1^2/\sigma_1^2}{s_2^2/\sigma_2^2}$$

- The F table gives percentiles of the F-distribution only for upper tail areas.

Estimation and Tests for Comparing Two Population Variances - cont.

- The lower tail values, when needed can be obtained by the relationship
$$F_{1-\alpha, df_1, df_2} = \frac{1}{F_{\alpha, df_1^*, df_2^*}}$$
where $df_1^* = df_2$ and $df_2^* = df_1$.
- That is, to calculate a left tail probability, look-up the corresponding right tail probability after switching the numerator and denominator degrees of freedom.
- For example $F_{.95,3,10}$ is calculated using the above relationship as $1/F_{.05,10,3} = 1/8.79 = 0.11$
- And $F_{.95,10,3}$ is calculated using the above relationship as $1/F_{.05,3,10} = 1/3.71 = 0.27$

Tests comparing two variances

Test:

1. $H_0 : \sigma_1^2 \leq \sigma_2^2$ vs. $H_a : \sigma_1^2 > \sigma_2^2$
2. $H_0 : \sigma_1^2 = \sigma_2^2$ vs. $H_a : \sigma_1^2 \neq \sigma_2^2$

Test Statistic:

$$F = \frac{s_1^2}{s_2^2}$$

Rejection Region: For given α, $df_1 = n_1 - 1$, and $df_2 = n_2 - 1$.

1. Reject H_0 if $F \geq F_{\alpha, df_1, df_2}$
2. Reject H_0 if $F \leq F_{1-\alpha/2, df_1, df_2}$ or $F \geq F_{\alpha/2, df_1, df_2}$

Example

In order to test the hypothesis $H_0 : \mu_1 - \mu_2 = 0$ using a two-sample t-statistic with pooled sample variance, we need to assume $\sigma_1^2 = \sigma_2^2$.

To check whether this is a valid assumption, let us formally test

$$H_0 : \sigma_1^2 = \sigma_2^2 \ vs. \ H_a : \sigma_1^2 \neq \sigma_2^2$$

using $\alpha = .05$

The two independent samples gave sample variances:

$s_1^2 = 0.105, \quad s_2^2 = 0.058$ with $df_1 = 9$ and $df_2 = 9$.

Example - cont.

Test: $H_0 : \sigma_1^2 = \sigma_2^2$ *vs.* $H_a : \sigma_1^2 \neq \sigma_2^2$

Test Statistic: $F_c = \frac{s_1^2}{s_2^2} = \frac{0.105}{0.058} = 1.81$

Rejection Region: $F \leq F_{.975,9,9}$ or $F \geq F_{.025,9,9}$

From Table 8, using $\alpha = 0.05$, $df_1 = 9$, $df_2 = 9$, we have

$F_{0.025,9,9} = 4.03$ and $F_{0.975,9,9} = 1/F_{0.025,9,9} = 1/4.03 = 0.25$

Thus we reject if $F \leq 0.25$ or $F \geq 4.03$.

Since $F_c = 1.81$ does not fall in the rejection region, we fail to reject $H_0 : \sigma_1^2 = \sigma_2^2$.

Conclusion: The assumption of equal variances for the two populations appears reasonable.

Test comparing several variances

Hartley's F_{max} Test for Homogeneity of Variances

This test requires that independent random samples of equal samples sizes, n, are drawn from t populations having normal distributions.

Test: $H_0 : \sigma_1^2 = \sigma_2^2 = \ldots = \sigma_t^2$ vs. H_a : Not all σ^2's equal

Test Statistic: $F_{max} = \frac{s^2_{max}}{s^2_{min}}$ where s^2_{max}=largest sample variance and s^2_{min}=smallest sample variance

Rejection Region: For given α, reject H_0 if F_{max} exceeds the tabulated percentile from Table 12, for α .

Note: Use t, $df_2 = n - 1$ and $a = \alpha$ to read Table 12

Test comparing several variances cont.

- When sample sizes are not all equal an approximate test is obtained if n is replaced with n_{max}, where n_{max} is largest sample size, in the above procedure.
- Hartley's test is quite sensitive to violation of the normality assumption.
- Thus other procedures that do not require the normality assumption, such as Levene's test has been proposed.
- Levene's test is too cumbersome to calculate by hand; so software may have to be used.
- Levine's test procedure is less powerful than Hartley's test when populations have normal distributions.